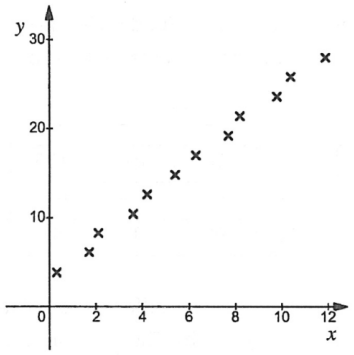
# Bivariate Data: PMCC and Linear Regression

Questions 1–25

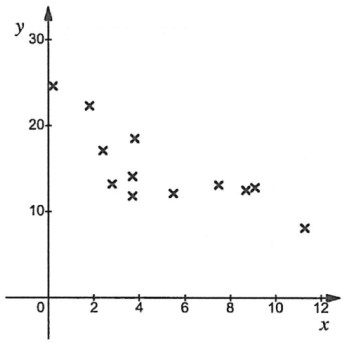
**Question 1.** The diagram below shows bivariate data.




Which of the following best describes the correlation shown in the diagram?

| | |
|---|---|
| Strong Positive | Moderate Positive |
| Weak Positive | Weak Negative |
| Moderate Negative | Strong Negative |

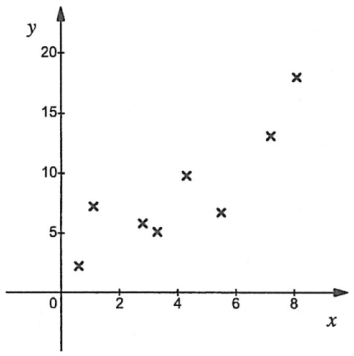
**Question 2.** The diagram below shows bivariate data.




Which of the following best describes the correlation shown in the diagram?

| | |
|---|---|
| Strong Positive | Moderate Positive |
| Weak Positive | Weak Negative |
| Moderate Negative | Strong Negative |

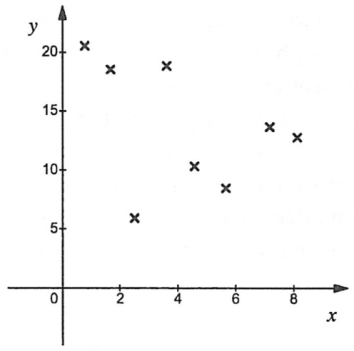
**Question 3.** The diagram below shows bivariate data.




One of the following product moment correlation coefficient is correct for the above data. Which is it?

$$\begin{array}{cc} 0.864 & 0.164 \\ -0.164 & -0.864 \end{array}$$

**Question 4.** The diagram below shows bivariate data.




One of the following product moment correlation coefficient is correct for the above data. Which is it?

$$\begin{array}{cc} 0.891 & 0.391 \\ -0.391 & -0.891 \end{array}$$

**Question 5.** Which of the following cannot be a value for the product moment correlation coefficient?

$$0 \qquad \frac{3}{4} \qquad -\frac{7}{3} \qquad -1 \qquad \frac{8}{7}$$

**Question 6.** A researcher conducts a study to understand the impact of physical exercise on mental health among college students. The researcher collects data on the number of hours students exercise per week and their levels of stress, measured on a standardised scale. Identify which of these is the explanatory variable and which is the response variable:

| | |
|---|---|
| Age of students | Hours of exercise per week |
| Diet quality | Levels of stress |

**Question 7.** Asher collects data on a person's age and their reaction time as measured in a standardised test. He calculates the product moment correlation coefficient to be 0.79. Which of the following best describes the pmcc found?

Definitely Correct Probably Correct
Probably Incorrect Definitely Incorrect

**Question 8.** Jane wants to determine if there is any correlation between a person's salary and how much they spend at a large supermarket. Jane collects a random sample of adults and calculates the product moment correlation coefficient to be 1.13. Which of the following best describes the pmcc found?

Definitely Correct Probably Correct
Probably Incorrect Definitely Incorrect

**Question 9.** Franklin has two data points: $(13, 7)$ and $(20, 11)$. He calculates the product moment correlation coefficient between $x$ and $y$ to be 1. Which of the following best describes the pmcc found?

Definitely Correct  Probably Correct
Probably Incorrect  Definitely Incorrect

**Question 10.** Ayaan works in an office in a busy city centre. He collects a random sample of his colleagues, and his results are shown in the table below.

| Distance from workplace (miles) | Commute Time (minutes) |
|---:|---:|
| 1.2 | 14 |
| 2.5 | 21 |
| 4.1 | 24 |
| 5.8 | 31 |
| 7.3 | 36 |
| 9.0 | 39 |
| 10.7 | 44 |
| 12.4 | 51 |

Use your calculator to calculate the product moment correlation coefficient between the distances from the workplace and the commute times.

**Question 11.** Grace asks a random sample of her the students in her year group how many hours they studied in a week (to the nearest hour) in preparation for a test, and their test score out of 100. The results are shown in the table below.

| Hours studied in a week | Test Score (/100) |
|---|---|
| 3 | 65 |
| 6 | 75 |
| 9 | 77 |
| 4 | 62 |
| 8 | 80 |
| 10 | 85 |
| 3 | 57 |
| 7 | 78 |

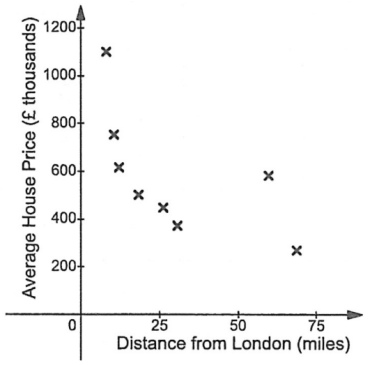
a) Use your calculator to calculate the product moment correlation coefficient between the hours studied in a week by the students and their test score.
b) Describe what the PMCC shows in this case, in context.

**Question 12.** A scientist is investigating the effects of smoking on the respiratory system in youths. A random sample of 15-18 year olds found a PMCC of $-0.76$ between the frequency of smoking and their respiratory function. Describe what this shows in context.

**Question 13.** Angela is investigating the average price of a house compared with the distance the house is away from London. She decides to consider areas in a South-Westerly direction from the centre of London. Her results are shown in the table below.

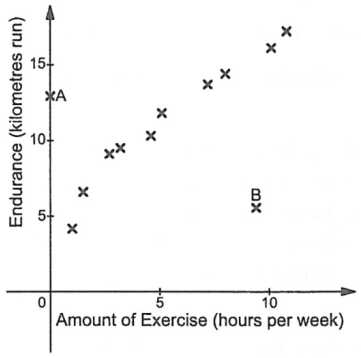
| Location | Dist. from London (miles) | Av. House Price (£thousands) |
|---|---|---|
| Richmond | 8.1 | 1100 |
| Twickenham | 10.5 | 753 |
| Hampton | 12.1 | 616 |
| Chertsey | 18.3 | 502 |
| Bagshot | 26.2 | 448 |
| Farnborough | 30.7 | 372 |
| Winchester | 59.8 | 582 |
| Tidworth | 68.7 | 269 |

**Question 13 (continued).** Angela plotted the data in a scatter graph, as shown below.



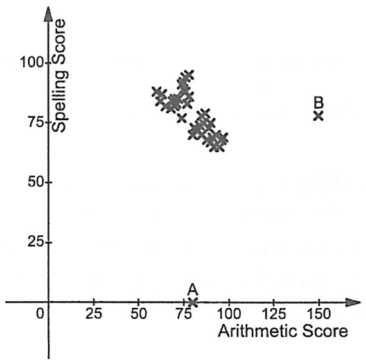

a) Identify a possible outlier and give a possible reason for it being an outlier.
b) If the outlier is discarded, determine if the scatter diagram shows any correlation for the remaining points.
c) Interpret the scatter diagram in context.

**Question 14.** Charlene organises a running club. In week 10, she plots the data of her running club members with the amount of exercise they do the previous week in hours along the horizontal axis, and their endurance, measured by how far they can run on the day they meet in kilometres along the vertical axis, as shown in the graph below.



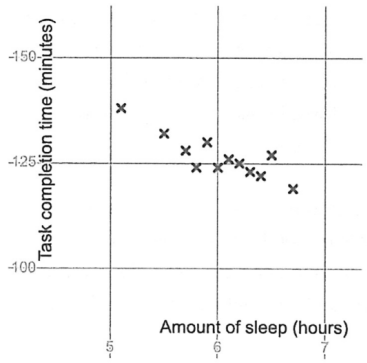

a) Explain why data point A may not have been inputted correctly.
b) Explain why data point B may have been inputted correctly.

**Question 15.** A teacher plots the scatter diagram below showing the arithmetic and spelling scores from a recent test sat by their students.



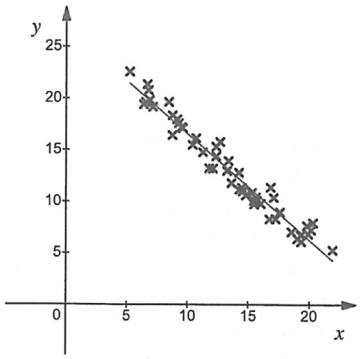

a) Explain why data point A may have been inputted correctly.
b) Explain why data point B may not have been inputted correctly.

**Question 16.** Tristan is doing some research into how sleep cycles affect productivity.
In his trial, he decides to randomly sample twelve adults and when they wake up they must complete a task.
The results of his experiment are shown in the diagram below.




Tristan claims that the scatter diagram proves that increasing the number of hours of sleep reduces the amount of time it takes to complete the task.
Comment on the validity of this claim.

**Question 17.** The diagram below shows bivariate data.




The product moment correlation coefficient for the data is $-0.979$. Explain why it would be reasonable to model the relationship between $x$ and $y$ as a linear graph.

**Question 18.** Maria wants to compare students' scores from a practice paper, where the maximum score was 100, against their total final exam score, where the maximum score was 300. She takes a random sample of students and the data is shown in the table below.

| Score from Practice Paper (/100) | Total Final Exam Score (/300) |
|---|---|
| 72 | 217 |
| 55 | 168 |
| 83 | 267 |
| 46 | 121 |
| 79 | 202 |
| 63 | 199 |
| 88 | 243 |
| 91 | 265 |

Use your calculator to calculate the equation of the regression line, where $x$ is the score on the practice paper and $y$ is the total final exam score.

**Question 19.** Chris is investigating how the amount of debt an adult has affects their credit score. He takes a random sample of adults with debts and collects data on the amount of debt they have, in thousands of pounds, and their credit scores. The data is shown in the table below.

| Amount of Debt (in thousands of dollars) | Credit Score |
|---|---|
| 12 | 723 |
| 18 | 716 |
| 21 | 663 |
| 27 | 668 |
| 37 | 672 |
| 38 | 650 |
| 41 | 565 |
| 49 | 575 |

Use your calculator to calculate the equation of the regression line, where $x$ is the amount of debt and $y$ is the credit score.

**Question 20.** The regression line between a company's monthly advertising expenditure (£$a$) and its monthly sales revenue (£$s$) is given by $s = 1.96a + 2920$. Describe in context what the 1.96 represents.

**Question 21.** A company makes cans of soup that are sold at different price points across the country. They want to investigate the relationship between how much the soup costs and how much they sell. The regression line between the price of a can of soup ($p$ pence) and the number of cans sold in a week ($N$) is given by $N = 502 - 2.22p$. Describe in context what the $-2.22$ represents.

**Question 22.** Qaisn is investigating how sleep affects sprinters' performance. He takes a random sample of runners and asks them how much sleep they had the night before, in hours, and then records their times running 100 metres. The regression line between the amount of sleep ($s$ hours) and their running times ($t$ seconds) is given by $t = 12.3 - 0.15s$. Describe a limitation with this model.

**Question 23.** Throughout the year, Peter runs a drop-in lunchtime workshop for his students that works on exam questions and exam technique. He maintains a register throughout the year and records the total number of hours each student attended, and their scores in the final exam (which is out of a maximum score of 300). The regression line between the number of hours they attended the workshop ($t$) and their final exam score ($q$) is given by $q = 98 + 1.12t$.

a) State the units of the value 1.12 in the regression line equation.

b) With reference to the regression line, describe what effect, on average, attending 2.5 hours of the lunchtime workshop throughout the year would have on a student's final exam score.

The one-hour lunchtime workshop runs Monday to Friday every week throughout the academic year for 38 weeks.

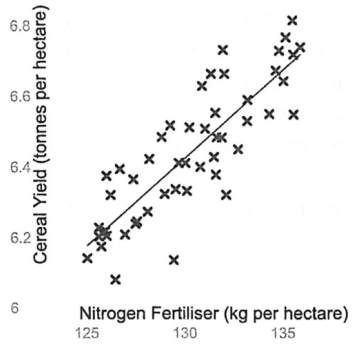
c) Peter wants to use the model to estimate the value of $q$ when $t = 200$. Explain why this would not be appropriate.

**Question 24.** Amelia is investigating the growth of a fungus. She measures the width of the fungus ($P$ mm) every twelve hours ($t$). She models the growth of the fungus using the regression line given by the equation $P = 3.1 + 0.67t$.

a) With reference to the regression line, describe what effect, on average, three days will have on the width of the fungus.

b) Assuming that the model continues to be accurate, use the regression line to determine an estimate for how many days it would take the fungus to reach a width of 5 cm.

**Question 25.** A random sample of farms report the amount of Nitrogen Fertiliser (kilogram per hectare) they used and their Crop Yield (tonnes per hectare). The data is shown in the diagram below.




The equation of the regression line is $y = 0.0497x - 0.0277$, where $y$ is the cereal yield and $x$ is the amount of nitrogen fertiliser.

a) Farm A reports that it uses 132 kg per hectare of nitrogen fertiliser. Use the equation of the regression line to estimate Farm A's cereal yield.

b) Farm B reports that it uses 121 kg per hectare of nitrogen fertiliser. Use the equation of the regression line to estimate Farm B's cereal yield.

c) Explain why the estimate for Farm A's cereal yield is more reliable than the estimate for Farm B's cereal yield.